WireSpot Technical Manual

Technical manual for operators, technicians, and support teams

Vexel Innovations | in collaboration with TechNova Technologies

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Product: WireSpot hotspot operations platform

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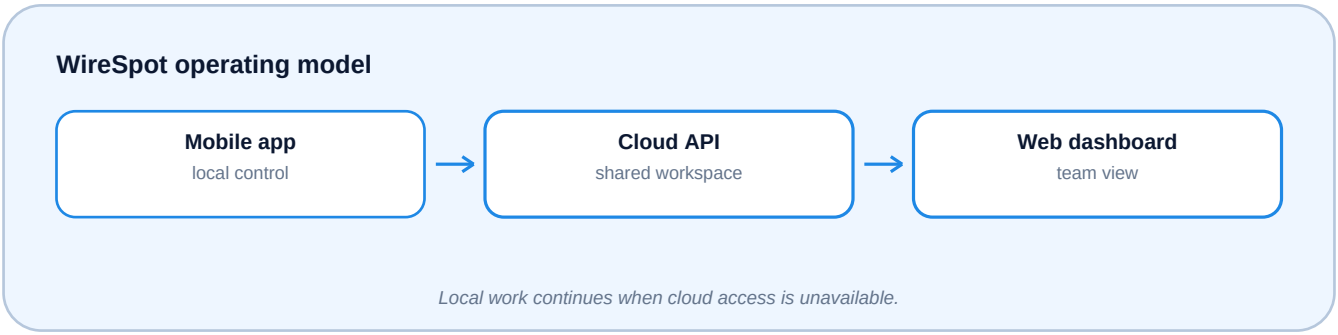
Audience: hotspot operators, field technicians, small ISPs, hotels, estates, and support teams

1. What WireSpot is



WireSpot brings router management, hotspot operations, vouchers, sales reports, printing, secure remote access, and cloud synchronization into one workspace. The Android application remains useful on a local network when the internet is unavailable. The hosted dashboard gives an authorized team a shared view of the same organization.

The current full live router connector is MikroTik RouterOS. Other vendors have safe connection or discovery foundations, but WireSpot deliberately refuses to invent CPU, memory, interface, hotspot, or command results where a vendor API mapping is not yet complete.

2. Before you start

Prepare:

- An Android phone for the first installation and field work.
- A MikroTik router running RouterOS with API access enabled.
- A local LAN connection, or a private tunnel such as WireGuard, Back To Home, or ZeroTier for remote management.
- An organization account and the same organization ID on each paired device.
- Optional: a Bluetooth ESC/POS printer, FreeRADIUS host, and a cloud API URL.

For remote production use, do not expose RouterOS management ports directly to the public internet. Use a private tunnel and firewall rules.

3. Install and secure the mobile app

1. Install the signed APK supplied by the organization or install the debug APK during field testing.

2. On first launch, create a local PIN. This PIN protects the device and is not the cloud access token.

3. Enable fingerprint, face, iris, Touch ID, Face ID, or Windows Hello when the platform offers it. The PIN remains the recovery fallback.

4. Follow the first-run walkthrough. It can be replayed from Settings.

5. Choose a theme, language, currency, and business identity.

The app stores router credentials in secure platform storage and scopes cloud router secrets by organization and router ID. Never paste a router password or operator token into a public issue, screenshot, or source file.

4. Add the first MikroTik router

On the router, check the API service:

```
/ip service print where name~"api"
/ip service enable api
```

Use API-SSL (8729) only after certificates and firewall rules are configured.

The standard API port is normally 8728.

In WireSpot:

1. Open **Routers** and choose **Add router**.

2. Select **MikroTik RouterOS**.

3. Enter a clear name, host/IP, API port, username, and password.

4. Select the access mode: **Local LAN**, **WireGuard**, **Back To Home**, **ZeroTier**, or another approved private path.

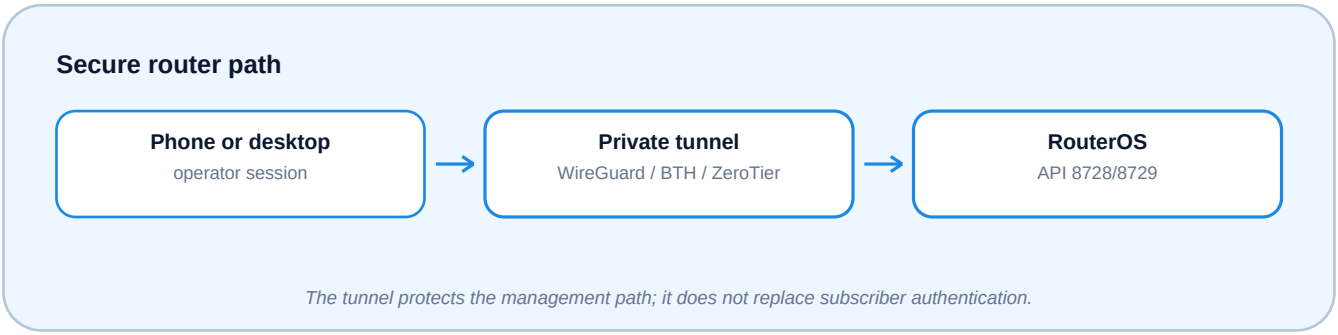
5. Save, then run **Test connection**.

6. Open the router card to inspect identity, health, interfaces, users, and available actions.

If the router is remote, connect the tunnel before testing the RouterOS API.

Local LAN mode is appropriate when the phone and router are on the same network.

5. Remote access and VPNs



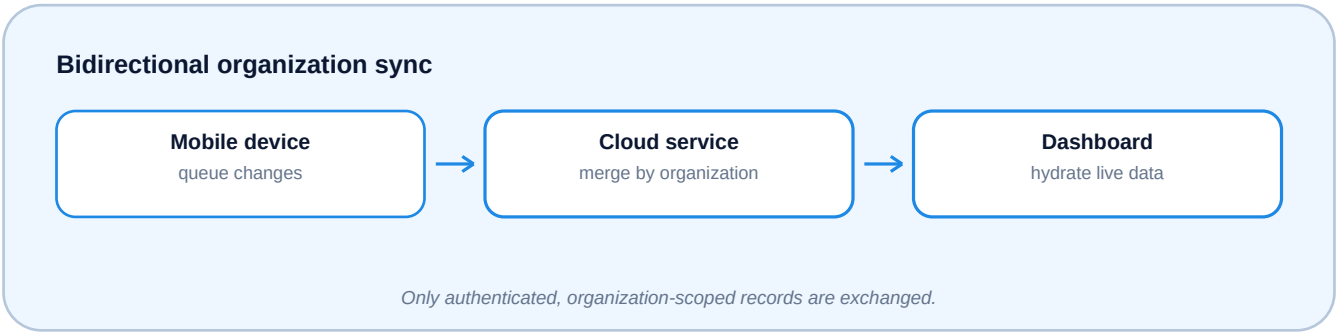
WireGuard is the preferred private management path. Import a configuration file

or scan its QR code from **Settings > WireGuard VPN**, approve the Android VPN permission, connect, and then test the router.

The app can store multiple profiles, but Android and iOS normally permit one active full-device VPN tunnel at a time. Use one profile with multiple peers when several router networks must be reachable simultaneously. A VPN provides the management path; it does not replace PPPoE subscriber authentication.

Back To Home and ZeroTier can be recorded as access modes when they provide the private route to the router. Public API-SSL is an advanced option and must be restricted by RouterOS firewall rules and trusted source addresses.

6. Cloud account and synchronization



The hosted dashboard is available at `https://wirespot.web.app/`. The reference API is `https://wirespot-api.onrender.com`.

1. Create an account or sign in on the dashboard.
2. In **Cloud Sync**, enter the API base URL and organization ID supplied by your administrator.
3. Sign in or pair the mobile device with the one-time pairing code shown in the activated phone's cloud session.
4. Test the connection, then choose **Sync now**.

The mobile app uploads organization-scoped router inventory, vouchers, settings, reports, printers, VPN profiles, groups, and telemetry frames. The dashboard hydrates its cards from the authenticated API and merges router inventory from multiple devices instead of replacing a fleet with the last writer. Changes made offline stay in the local queue and are retried after connectivity returns.

The web and mobile clients must use the same organization ID. A successful `/health` request only proves that the API is reachable; it does not prove that the account is authenticated or that a router is online.

Production backend note

The hosted reference service currently reports development-token mode until its

operator config is changed. A production deployment must set:

- `WIRESPOT_ENV=production` (or `WIRESPOT_PRODUCTION=true`)
- `WIRESPOT_STORE_PATH` on persistent storage, or replace the JSON store with a managed database/Firestore implementation
- either `WIRESPOT_OPERATOR_TOKENS` with server-owned entries in the form `token|organizationId|role`, or the explicit `WIRESPOT_ENABLE_LOCAL_AUTH=true` switch for a single-instance pilot

The local-auth switch keeps the built-in account login and registration usable for a controlled pilot. It is not a replacement for a hosted identity provider in a public multi-tenant deployment. Free sleeping hosts are fine for pilots but do not guarantee uptime or durable storage. Run the hosted smoke test with `WIRESPOT_REQUIRE_PRODUCTION=true`; add a dedicated smoke account through `WIRESPOT_SMOKE_EMAIL` and `WIRESPOT_SMOKE_PASSWORD` to verify authenticated router reads, and optionally `WIRESPOT_SMOKE_ROUTER_JSON` to verify a complete write/read sync round-trip.

7. Hotspot, subscribers, PPPoE, and RADIUS

The MikroTik hotspot screens manage users, profiles, active sessions, queues, cookies, IP bindings, disconnects, counters, and deletion. Voucher generation supports batches, PIN-style credentials, QR payloads, plans, prices, and expiry.

For PPPoE subscriber access, deploy the bundled FreeRADIUS service on a private host. Configure RouterOS with the RADIUS address, UDP ports `1812` and `1813`, and a strong shared secret. PPPoE/RADIUS authenticates subscribers and records accounting; it is separate from the secure cloud management path. Do not expose RADIUS UDP ports to the public internet.

The starter RADIUS policy is suitable for a lab. Production deployments should move subscriber data to SQL or a protected REST-backed policy service.

8. Reports, vouchers, and printing

Reports support daily, weekly, and monthly views with revenue and sales detail. Export CSV for spreadsheets, PDF for sharing or filing, and JSON for backups or integration. Voucher receipts support QR codes, co-branding, and 58 mm or 80 mm ESC/POS layouts.

To print, pair the Bluetooth printer in the operating system, add it under **Settings > Printers**, choose the paper width, and print a preview receipt.

Real printer testing is required because ESC/POS support differs by model.

9. Roles and safe operations

Organizations use owner, admin, manager, operator, and viewer roles. The server checks organization and role claims; the app also hides or disables actions that the current role cannot perform. Treat the viewer role as read-only.

High-impact operations such as router commands, reboots, backup restore, and license administration require an appropriate role. Review the target router and command before confirming. Keep tokens, pairing codes, and credentials private.

10. Vendor support status

Vendor	Current support
MikroTik RouterOS	Live connection, snapshots, hotspot/voucher operations, and command execution
Ruijie / Reyee	Authenticated Ruijie Cloud connection, device discovery, and read-only dashboard snapshot
OpenWrt	Authenticated read-only ubus dashboard snapshot; mutations pending
TP-Link Omada	Authenticated controller site/device reads and read-only dashboard snapshot; mutations pending
Ubiquiti UniFi	Authenticated Network Integration API site/device snapshot; normalized mutations pending
Generic	Connection/probe foundation only

Unsupported vendor commands and snapshots fail clearly. They are not replaced by made-up telemetry. Use the vendor's own controller until a tested WireSpot adapter is released.

11. Windows, Linux, and iOS

Windows and Linux use the Flutter desktop application and share the cloud and local workflows. Windows can use WireGuard for Windows, Windows printer queues, and Windows Hello where installed. Distribute the complete release folder, not only the executable, because Flutter DLLs and data files are required.

The iOS target requires Apple signing and a macOS/Xcode workflow for a real device or TestFlight. Unsigned CI output is for build validation only. Native WireGuard, Bluetooth printing, and biometric behavior must be verified on an iPhone before release.

Companion devices use the same organization subscription. A desktop can be paired to an activated phone with a one-time pairing code instead of requiring a second subscription; device limits and entitlement policy remain server-owned.

12. Troubleshooting

Cloud backup returns 404. Confirm the API base URL, organization ID, and authenticated session. A healthy API does not imply that a backup exists yet.

The dashboard shows a router offline. Refresh the dashboard, verify that the mobile agent is authenticated for the same organization, check the router's local/VPN path, and run Test connection. Telemetry is only marked live when a real connector has delivered a recent snapshot.

A non-MikroTik command is rejected. This is intentional until that vendor's command mapping is implemented and tested. Use the vendor controller.

The Android APK reports `libsqlite3.so` missing. Install a current artifact from the CI build after the native SQLite library bundling fix; do not reuse an older APK cached on the phone.

Cloud sync appears empty. Check that both devices use the same organization ID and account, then run Sync now. Inspect pending operations and the last sync time before retrying.

Router ping fails. Ping tests the management path, not just internet access. Confirm the tunnel is connected, the host/port is reachable from that tunnel, RouterOS API is enabled, and the credentials have the required policy.

13. Support and release checklist

Before a field rollout:

1. Run `flutter analyze` and `flutter test`.
2. Test a real MikroTik on LAN and through the chosen VPN path.
3. Create a voucher, print a receipt, export a CSV/PDF, and restore a backup.
4. Verify the same organization appears on mobile and web.
5. Confirm role restrictions with an operator and viewer account.
6. Confirm production backend configuration and persistent storage.
7. Record the deployed app version and keep a rollback APK.

This manual is intended to be self-contained. If support is needed, use the contact details in the app's About page and include the app version, device, router vendor, connection mode, and the exact error message. Never include passwords, access tokens, pairing codes, or customer data in a support request.

14. Product identity

WireSpot is a Vexel Innovations product developed in collaboration with TechNova Technologies. Product support, licensing, and release information

should use the approved contact details shown in the app's About page and the organization's published support channel.

15. Quick reference and glossary

API Base URL: The HTTPS address of the WireSpot cloud service. It is not the router address and should never be replaced with a private RouterOS IP.

Organization ID: The workspace identifier shared by the mobile app and web dashboard. Devices using different organization IDs cannot see one another's routers, vouchers, or reports.

Access token: A secret bearer credential used for cloud API requests. It is created by the configured sign-in or pairing flow and should be treated like a password. It is not the local PIN, RouterOS password, or RADIUS secret.

Pairing code: A short-lived, one-time code displayed by an activated device to authorize a companion desktop or web session.

Telemetry: A time-stamped observation obtained from a real router connector. When a connector cannot provide a value, WireSpot leaves it unavailable rather than displaying a guessed number.

RouterOS API: MikroTik's management protocol. Port `8728` is the usual plain API port; `8729` is the usual API-SSL port.

RADIUS: The authentication and accounting service used for subscriber access such as PPPoE. RADIUS is not a VPN and does not make a router remotely reachable by itself.

VPN management path: A private route, such as WireGuard, Back To Home, or ZeroTier, through which the app can reach a router without publishing its management port.

Daily operator checklist

1. Confirm the correct organization and current sync time.
2. Check that the active router has a recent real snapshot.
3. Review active sessions and sales before issuing vouchers.
4. Export or back up important records before major configuration changes.
5. Disconnect the VPN when field work is complete if it is not needed.
6. Sign out of shared computers and never leave tokens in screenshots.